

IDEA SUBMISSION

Theme — [Fill from portal]

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Problem Statement Category — [Fill from portal]

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RTMTS Scanner

Real-Time Threat Monitoring System · powered by the SentinelAI engine

Detects the threat. Scores the risk. Explains itself — in real time.

IDEA TITLE

RTMITS Scanner — a real-time, source-agnostic threat-detection platform that explains every alert.

WHAT IT IS

One engine watches every activity stream — payments, logins, database queries, blockchain wallets. It learns each entity's normal behavior and flags deviations in real time with a 0–100 risk score and a plain-English reason. A working prototype — 63 backend tests passing.

HOW IT SOLVES THE PROBLEM

Legacy tools fire static-threshold alerts and dump raw rows with no context, so analysts drown in noise and catch attacks too late. We use per-entity behavioral baselines (catching novel attacks) and pre-explain every alert — triage drops from minutes to seconds.

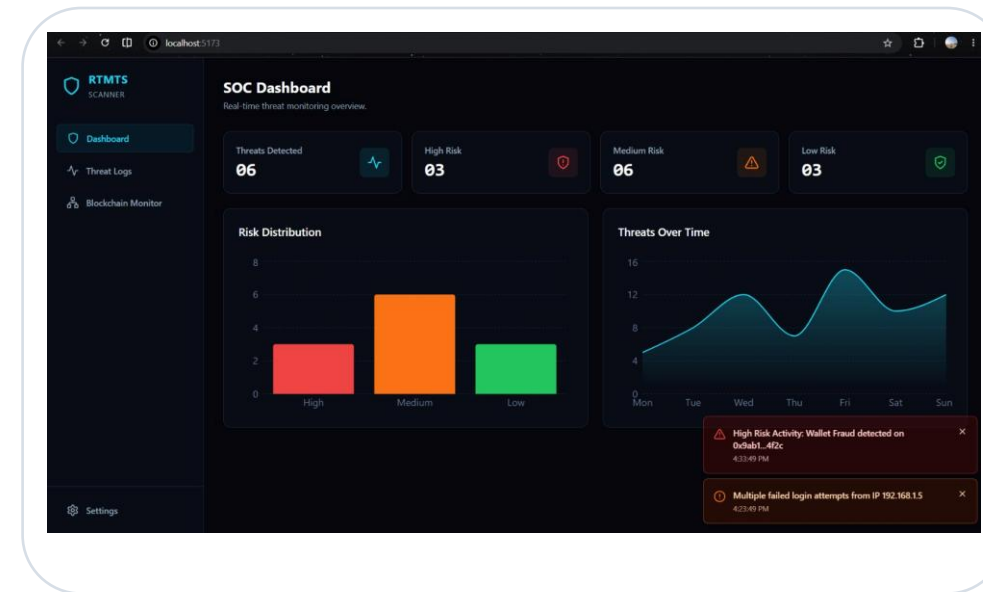
WHAT MAKES IT NOVEL

- Source-agnostic — a new data source = one small adapter file. A platform, not a dashboard.
- Explainable by default — Gemini turns an anomaly into one human sentence (cached, offline-safe).
- Two-layer detection — statistical spike + ML drift fused into one intuitive score.

63 tests passing

Offline-safe demo

Source-agnostic core



LIVE SOC DASHBOARD

Press “inject attack” → within ~1–2s the entity is flagged, scored, and explained — every run, offline-safe.

SOLUTION APPROACH

TECHNOLOGY STACK

React · Tailwind · Recharts

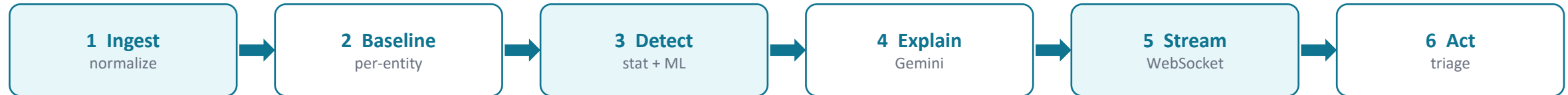
FastAPI · WebSockets · Python

scikit-learn IsolationForest

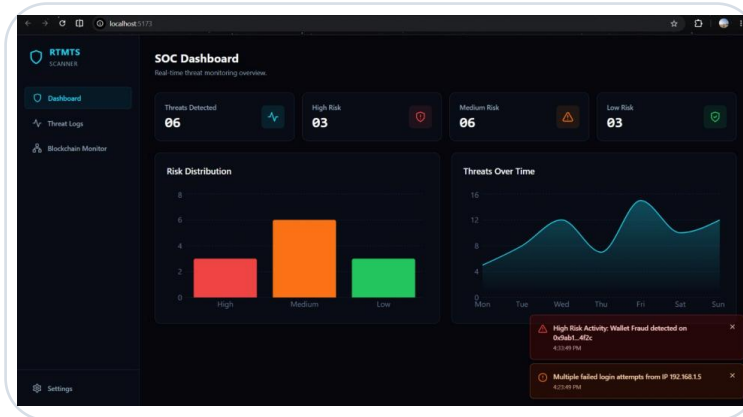
Google Gemini

Etherscan / Alchemy

DETECTION PIPELINE



WORKING PROTOTYPE



SOC Dashboard

The Threat Logs section displays a detailed view of all detected security events. It includes a table with columns for ID, Threat Type, Score, Timestamp, and Status. The table lists several events, including Wallet Fraud, DB Attack, DDoS, Brute Force, Unauthorized Access, and Phishing Attempt. Each event has a corresponding score and a status (Active, Investigating, Resolved). At the bottom, there are two alert notifications: 'High Risk Activity: Wallet Fraud detected on 0x8a1...402c' and 'Multiple failed login attempts from IP 192.168.1.5'.

ID	Threat Type	Score	Timestamp	Status
THR-001	Wallet Fraud	92	6/23/2024, 4:28:49 PM	Active
THR-002	DB Attack	87	6/23/2024, 4:03:49 PM	Investigating
THR-003	DDoS	82	6/23/2024, 11:33:49 AM	Investigating
THR-004	Brute Force	76	6/23/2024, 1:13:49 PM	Investigating
THR-005	Unauthorized Access	65	6/23/2024, 3:33:49 PM	Resolved
THR-006	Phishing Attempt	48	6/23/2024, 2:33:49 PM	Active

Threat Logs — scored events

The Blockchain Monitor section provides real-time tracking of suspicious wallet activities and transactions. It includes a table with columns for Wallet Address, Amount (ETH), Status, and Actions. The table lists several transactions, including a high-risk activity detected on 0x8a1...402c. At the bottom, there are two alert notifications: 'High Risk Activity: Wallet Fraud detected on 0x8a1...402c' and 'Multiple failed login attempts from IP 192.168.1.5'.

Wallet Address	Amount (ETH)	Status	Actions
0x472a...472a	11.2800	Normal	View Source
0x0a1b...472c	419.5000	Flagged	View Source
0x1111...2222	12.0000	Normal	View Source
0x3333...4444	10000.0000	Resolved	View Source
0x5555...6666	200.0000	Flagged	View Source

Blockchain Monitor

FEASIBILITY AND VIABILITY

FEASIBLE TODAY

- Already built & demoable — live dashboard, real-time engine, attack-on-cue, plain-English explanations, 63 passing tests.
- Lean 2-service architecture (React ↔ FastAPI) — runs on a single laptop, no cloud dependency in the demo.
- Built entirely on proven, free open-source components — no exotic or costly infrastructure.

RISK → MITIGATION

Live demo on bad wifi

Gemini cached + template fallback; binds localhost — fully offline-safe.

ML cold-start (needs history)

Warm-up seeding builds baselines; fast statistical layer carries the live spike.

False positives

Per-entity baselines + 2-layer fusion; never learns from confirmed anomalies.

Scaling to real volume

Stateless engine shards per entity; in-memory store is swappable.

Securing the product itself

OWASP self-audit (SECURITY.md): no critical or high findings.

IMPACT AND BENEFITS

Target audience: banks & fintechs · SOC / security teams · crypto exchanges & wallets · any org with high-volume activity streams.

~1–2s

from attack to flagged, scored & explained

1 adapter

to onboard a brand-new data source

0

critical / high security findings (OWASP audit)

BENEFITS

Economic — Cuts fraud losses, chargebacks & breach-remediation cost; fewer analyst-hours per alert; one platform replaces several point tools.

Social / trust — Protects customers' money and data; faster response means less harm; supports safer digital finance.

Operational — One engine for all sources = lower integration & maintenance cost; new source onboarded with a single adapter.

Efficiency — Lean architecture with an in-memory hot path = low compute footprint vs. heavy traditional SIEM stacks.

RESEARCH AND REFERENCES

- **Isolation Forest** — Liu, Ting & Zhou — “Isolation Forest,” IEEE ICDM 2008 (core anomaly-detection algorithm).
- **scikit-learn** — IsolationForest & outlier-detection documentation — scikit-learn.org.
- **UEBA** — User & Entity Behavior Analytics — per-entity behavioral baselining (industry concept).
- **OWASP Top 10 (2021)** — Framework used for our security self-audit — owasp.org/Top10.
- **Core stack docs** — FastAPI · React · Recharts · Tailwind CSS — official documentation.
- **Google Gemini API** — ai.google.dev — explanation layer.
- **Etherscan / Alchemy APIs** — blockchain wallet-activity module (optional).
- **Validation roadmap** — Public fraud datasets (IEEE-CIS / credit-card fraud, Kaggle) for benchmarking.